SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA0304	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	30% of CIA		Slot B
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	AI & ML	**Date:**	10-08-2026

Code of Conduct

I G.V.S.HEMANTH/192425186(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G.V.S.HEMANTH

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AIPowered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System
Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity	Case Studies 1, 2, and 3

Applications	resolution, and decision-making in real-world NLP systems	

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

"Data science is powerful, data science drives innovation, data science is evolving."

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ_1 = 0.5 (Trigram)
- λ_2 = 0.3 (Bigram)
- λ_3 = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now." 2.
   "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    ○ economic (120 occurrences) ○
    growth (450 occurrences) ○
    increases (210 occurrences) ○
    employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No.	Understandin	Application	Analysis	Solution	Clarity	Sub.	Total

/ Criteri a Level	g of Case	of Concepts		Design		Total	%
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _______________	Signature: _______________	Signature: _______________
Date: ____________________	Date: ____________________	Date: ____________________

Case Study 1 – Smart Mobile Keyboard Prediction System

The corpus is:

“Data science is powerful, data science drives innovation, data science is evolving.”

The given statistics and interpolation weights are provided in the assessment document.

Q1. Compute the MLE of the Bigram Probability P(science|data)

For a bigram model:

P(wi|wi−1)=C(wi−1)C(wi−1,wi)

Given:

C(data)=3 C(data science)=3

Therefore,

P(science|data)=C(data)C(data science) =33=1.0

Interpretation

The probability is **1.0 or 100%**, meaning that in the given training corpus, every occurrence of **"data"** is followed by **"science."**

Therefore, a mobile keyboard trained on this corpus would assign very high confidence to **"science"** when the user types **"data."** In a real keyboard, this would make *science* a strong next-word prediction, although a larger and more diverse corpus would be needed to generalize reliably.

The assessment specifically asks for the MLE and its effect on real-time next-word prediction.

Q2. Backoff Model for the Unseen Sequence "data science improves"

The sequence **"data science improves"** is unseen in the corpus. The assessment asks us to use lower-order n-grams through a Backoff Model.

For the trigram:

P(improves|data,science)

we have:

C(data,science,improves)=0

Hence the trigram probability is:

P(improves|data,science)=0

Backoff to the bigram

We next consider:

P(improves|science)

There is no occurrence of **"science improves"** in the corpus:

C(science,improves)=0

Therefore:

P(improves|science)=0

Backoff to unigram

The word **“improves”** does not occur anywhere in the supplied corpus. Therefore its observed unigram probability is also zero:

$P(\text{improves})=0$

Thus, **using only the supplied corpus statistics**, the estimated probability remains:

$P(\text{improves}|\text{data,science})=0$

Why is Backoff essential?

Backoff is important because real-world text contains many unseen word combinations. Instead of relying exclusively on a high-order trigram, a predictive system can fall back to a bigram or unigram model.

For example:

Trigram→Bigram→Unigram

This improves **coverage and robustness** when the training corpus is sparse. In this particular small corpus, however, even the lower-order models have no evidence for *improves*, so the supplied data cannot assign it a non-zero probability without additional smoothing or an external vocabulary/corpus.

Q3. Deleted Interpolation for “data science is”

The assessment provides:

$\lambda 1 = 0.5(\text{Trigram})\ \lambda 2 = 0.3(\text{Bigram})\ \lambda 3 = 0.2(\text{Unigram})$

These weights sum to 1:

$0.5+0.3+0.2=1$

The interpolation formula is:

$P(\text{is}|\text{data,science})=\lambda 1\ P(\text{is}|\text{data,science})+\lambda 2\ P(\text{is}|\text{science})+\lambda 3\ P(\text{is})$

Step 1: Trigram probability

The trigram **“data science is”** occurs twice.

$C(\text{data,science,is})=2$

and:

$C(\text{data,science})=3$

Therefore:

$P(\text{is}|\text{data,science})=32=0.6667$

Step 2: Bigram probability

Given:

$C(\text{science,is})=2$

and there are three occurrences of **science**:

$C(\text{science})=3$

Therefore:

$P(\text{is}|\text{science})=32=0.6667$

This is also consistent with the supplied prediction probability of approximately 0.66 after *science*.

Step 3: Unigram probability

The corpus contains 12 words in total, and **is** occurs twice.

$P(\text{is})=122=61=0.1667$

Step 4: Apply interpolation

$P(\text{is}|\text{data},\text{science})=(0.5)(0.6667)+(0.3)(0.6667)+(0.2)(0.1667)=0.33335+0.20001+0.03334$
$P(\text{is}|\text{data},\text{science})\approx 0.5667$

If the probability of the entire three-word sequence is required

We can additionally calculate:

$P(\text{data},\text{science},\text{is})=P(\text{data})P(\text{science}|\text{data})P(\text{is}|\text{data},\text{science})$

Since $P(\text{data})=3/12=0.25$ and $P(\text{science}|\text{data})=1$:

$P(\text{data},\text{science},\text{is})=0.25(1)(0.5667)\approx 0.1417$

For the interpolation question, **0.5667 is the interpolated next-word probability**; **0.1417 is the probability of the complete sequence** under this factorization.

Interpretation

Deleted interpolation combines:

- **Trigram information** — more context-specific and therefore given the largest weight.
- **Bigram information** — useful when trigram evidence is sparse.
- **Unigram information** — provides general word-frequency information.

Thus, interpolation provides a balance between **reliability and coverage**, reducing dependence on any single sparse n-gram model. The assessment explicitly asks for this interpretation.

Q4. Entropy of P(is)=0.66 and P(drives)=0.33

Entropy measures the uncertainty of a probability distribution:

$H(X)=-i\sum P(xi)\log 2P(xi)$

Given:

$P(\text{is})=0.66$ $P(\text{drives})=0.33$

Therefore:

$H(X) = -(0.66\log_2 0.66 + 0.33\log_2 0.33)$ $H(X) \approx 0.923$ bits

The probabilities provided in the question sum to 0.99 because of rounding. If normalized to 2/3 and 1/3, the entropy is approximately **0.918 bits**.

Interpretation

The entropy is relatively low compared with the maximum entropy of **1 bit** for two equally likely choices.

This indicates that the keyboard has **some uncertainty**, but the uncertainty is not maximal because:

$P(\text{is}) = 0.66 > P(\text{drives}) = 0.33$

Hence, the keyboard can confidently rank **"is"** above **"drives"** as the next-word prediction.

Role of entropy in a keyboard

Entropy helps measure how uncertain the prediction system is:

- **Low entropy** → prediction is concentrated on a few words → higher confidence.
- **High entropy** → many words have similar probabilities → greater uncertainty.
- The system can use this information to determine how confidently it should present its next-word suggestions.

Final Answers at a Glance

Question	**Answer**
Q1	$P(\text{science} \mid \text{data}) = 1.0$
Q2	Supplied corpus gives $P(\text{improves} \mid \text{data}, \text{science}) = 0$ even after backing off
Q3	Interpolated $P(\text{is} \mid \text{data}, \text{science}) = 0.5667$; complete sequence probability ≈ 0.1417
Q4	Entropy 0.923 bits using the supplied rounded probabilities

This format directly follows the four questions in **Case Study 1** of the assessment.

Case Study 2 – AI-Powered Customer Support Chatbot

The case study describes an airline chatbot that uses the **Penn Treebank POS tagset** and an **HMM** to determine whether the ambiguous word **"book"** is a verb or noun. The given probabilities are $P(\text{book} \mid VB) = 0.6$, $P(\text{book} \mid NN) = 0.4$, and $P(\text{Start} \to VB) = P(\text{Start} \to NN) = 0.5$.

Q1. Assign POS tags to both sentences

Sentence 1: "Book a flight ticket now."

Word	POS Tag	Reason
Book	**VB**	Used as an action/command
a	**DT**	Determiner
flight	**NN**	Noun modifying "ticket"
ticket	**NN**	Noun
now	**RB**	Adverb

Tagged sentence:

Book/VB a/DT flight/NN ticket/NN now/RB

Here, **book** means to reserve something, so it functions as a **verb (VB)**.

Sentence 2: "This book is interesting."

Word	POS Tag	Reason
This	**DT**	Determiner
book	**NN**	Refers to a physical/written object
is	**VBZ**	Third-person singular present form of "be"
interesting	**JJ**	Adjective

Tagged sentence:

This/DT book/NN is/VBZ interesting/JJ

Here, **book** is the subject/object being described, so it is a **noun (NN)**.

Contextual ambiguity

The word **book** has two possible POS tags:

$book \rightarrow \{VB, NN\}$

Its correct interpretation depends on its surrounding context. The assessment specifically asks for contextual justification of this ambiguity.

Q2. HMM likelihood of "book" being tagged as VB

The HMM considers both:

1. The probability of starting with the tag.
2. The probability of emitting the word from that tag.

Given:

$P(Start \rightarrow VB) = 0.5$ $P(book|VB) = 0.6$

Therefore:

$P(VB, book) = P(Start \rightarrow VB) \times P(book|VB)$ $= 0.5 \times 0.6$ $P(VB, book) = 0.30$

For comparison, for NN:

$P(NN, book) = P(Start \rightarrow NN) \times P(book|NN)$ $= 0.5 \times 0.4$ $P(NN, book) = 0.20$

Since:

$0.30 > 0.20$

the HMM favors **VB**.

Normalized probability

If we normalize the two alternatives:

$P(VB|book) = 0.30 + 0.200.30$ $= 0.60$

Similarly:

$P(NN|book) = 0.500.20 = 0.40$

Therefore, based **only on the probabilities supplied in the case study**, the model gives a **60% preference to VB** and **40% to NN**.

Why does it favor VB?

Because:

$P(book|VB) = 0.6$

is greater than:

$P(book|NN) = 0.4$

while both starting probabilities are equal at 0.5. Hence, the emission probability is what makes VB more likely.

Note: The question asks for the likelihood of the word being tagged VB using the given HMM probabilities. The calculation $0.5 \times 0.6 = 0.30$ is the direct HMM path score; 0.60 is the normalized comparison between the two supplied tag alternatives.

Q3. Rule-Based Tagging vs Stochastic (HMM) Tagging

Aspect	**Rule-Based Tagging**	**Stochastic / HMM Tagging**
Method	Uses manually designed linguistic rules	Uses probabilities learned from data
Ambiguity	Resolves using predefined contextual rules	Resolves using emission and transition probabilities

Data requirement	Less training data required	Requires a suitable tagged corpus
Flexibility	Limited when encountering new patterns	More adaptable to statistical patterns
Scalability	Difficult to maintain many rules	More suitable for large datasets
Example	"Book" after a command structure → VB	Compares probabilities of VB and NN

Recommendation

For a **large-scale chatbot deployment**, **stochastic/HMM tagging is more suitable** because it can use corpus-derived probabilities to handle contextual ambiguity and can be scaled across large amounts of language data.

However, rule-based methods can still be useful for **specific linguistic corrections or deterministic patterns**. A practical system can therefore combine statistical tagging with carefully designed rules.

The assessment explicitly asks for a comparison and a recommendation for large-scale chatbot deployment.

Q4. Role of Standardized POS Tagsets and Word Classes

The **Penn Treebank POS tagset** gives words standardized grammatical categories such as:

- **VB** – Verb
- **NN** – Noun
- **DT** – Determiner
- **RB** – Adverb
- **JJ** – Adjective
- **VBZ** – Verb, 3rd-person singular present

Standardization helps the chatbot understand the **syntactic structure** of user messages.

1. Intent Detection

POS information helps distinguish different meanings.

For example:

- **"Book a flight."** → book/VB → action/request
- **"This book is interesting."** → book/NN → object/entity

Thus, POS information helps the chatbot distinguish the user's intended action from a reference to an object.

2. Response Generation

Knowing whether a word is a verb, noun, adjective, etc. helps the chatbot construct grammatically appropriate responses and identify important entities and actions.

3. Ambiguity Resolution

A word can have multiple grammatical roles. POS tagging provides a structured representation that allows the chatbot to select the appropriate interpretation based on context and probabilities.

4. Overall Performance

Standardized word classes make NLP processing more consistent and allow downstream components such as **intent detection, semantic analysis, and response generation** to work with structured linguistic information.

Therefore:

POS Tagging→Better Syntax Understanding→Better Intent Detection→Better Responses

This aligns with the case study's focus on POS tagging, word classes, ambiguity resolution, and chatbot language understanding.

Final Answers

Question	**Answer**
Q1	Book/VB a/DT flight/NN ticket/NN now/RB; This/DT book/NN is/VBZ interesting/JJ
Q2	VB path score = **0.30**; normalized (P(VB
Q3	**HMM/Stochastic tagging** is preferable for large-scale deployment because it handles ambiguity probabilistically and scales better
Q4	Standard POS tags provide consistent grammatical information, improving **intent detection, ambiguity resolution, response generation, and overall chatbot understanding**

Case Study 3 – News Analytics and POS Tag Correction System

The case study uses a **Transformation-Based Tagger (Brill Tagger)** to correct POS tags after an initial tagging stage. The sentence given is **"Economic growth increases employment."** with the initial tags shown in the assessment.

Q1. Apply the transformation rule and update the POS tag sequence

Given sentence

Economic growth increases employment.

Initial POS tags

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

The learned transformation rule is:

Change NNS to VBZ if the preceding word is tagged as NN.

Here:

- **growth** → NN
- **increases** → NNS

Since the preceding word **growth** is tagged **NN**, the rule applies:

$NNS \rightarrow VBZ$

Therefore:

Corrected sequence

economic/JJ growth/NN increases/VBZ employment/NN

Why is this linguistically appropriate?

In the sentence:

Economic growth increases employment.

"growth" is the subject and is singular. Therefore, the following verb should agree with the singular subject:

$growth \rightarrow increases$

The word **"increases"** is a third-person singular present-tense verb, so **VBZ** is appropriate.

Thus, the transformation corrects the initial **NNS** tag to **VBZ**.

Q2. Analyze the initial tagging errors

The initial sequence is:

Word	Initial Tag	Corrected Tag	Explanation
economic	JJ	**JJ**	Describes/modifies "growth"
growth	NN	**NN**	Singular noun and subject

increases	NNS	**VBZ**	Verb agreeing with singular subject
employment	NN	**NN**	Noun functioning as the object

Error

The main tagging error is:

increases/NNS→increases/VBZ

NNS represents a plural noun, whereas **VBZ** represents a third-person singular present-tense verb.

The surrounding grammatical structure makes the intended interpretation clear:

singular subjectEconomic growthverbincreasesobjectemployment

Therefore, the corrected sentence is:

economic/JJ growth/NN increases/VBZ employment/NN

The assessment specifically asks the correction to be justified using both **grammatical structure and sentence semantics**.

Q3. Compute the word frequency distribution

The corpus frequencies given are:

- economic = **120**
- growth = **450**
- increases = **210**
- employment = **380**

Step 1: Calculate total frequency

N=120+450+210+380 N=1160

Step 2: Calculate probability of each word

The frequency distribution is:

P(w)=NC(w)

Word	**Frequency**	**Probability**	**Percentage**
economic	120	120/1160=0.1034	**10.34%**
growth	450	450/1160=0.3879	**38.79%**
increases	210	210/1160=0.1810	**18.10%**
employment	380	380/1160=0.3276	**32.76%**
Total	**1160**	**1.0000**	**100%**

Therefore, the word frequency distribution is:

$P(\text{economic})P(\text{growth})P(\text{increases})P(\text{employment})=0.1034=0.3879=0.1810=0.3276$

How does frequency support probabilistic POS tagging?

Frequency information provides evidence about how commonly words occur in a corpus. A probabilistic tagger can use such corpus statistics together with contextual information to estimate likely tags.

For example, **growth** has the highest frequency among these four words:

$P(\text{growth})=0.3879$

while **economic** has the lowest:

$P(\text{economic})=0.1034$

However, **frequency alone does not determine POS**. Context and grammatical relationships are also necessary. This is particularly important for ambiguous words.

Q4. How Transformation-Based Tagging improves accuracy

Transformation-Based Tagging works by:

Initial Tags→Find Error Pattern→Apply Learned Rule→Corrected Tags

In this case:

Initial tagging

economic/JJ growth/NN increases/NNS employment/NN

Transformation rule

NNS→VBZ if previous tag is NN

Final tagging

economic/JJ growth/NN increases/VBZ employment/NN

The transformation therefore removes the incorrect **NNS** assignment and produces a grammatically appropriate **VBZ** tag.

Advantages

1. **Corrects systematic tagging errors**
   The tagger identifies recurring contextual errors and applies learned rules.
2. **Uses context**
   Instead of considering a word independently, it examines neighboring tags.
3. **Improves linguistic consistency**
   The corrected sequence agrees with the grammatical structure of the sentence.
4. **Refines an initial tagger's output**
   The Brill Tagger can operate as a correction stage after an initial POS tagging process.

Entropy and uncertainty

Entropy can be used to measure uncertainty in POS assignments.

The general formula is:

$H(X)=-i\sum P(xi)\log 2P(xi)$

Suppose a word has two possible tags with similar probabilities:

$P(NN)=0.5,P(VBZ)=0.5$

Then:

$H(X)=-(0.5\log 20.5+0.5\log 20.5)=1$ bit

This represents **high uncertainty**.

After contextual evidence and the transformation rule strongly favor one tag, the distribution becomes more concentrated, so entropy decreases.

For example:

$P(VBZ)=0.9,P(NNS)=0.1$

gives:

$H(X)\approx 0.469$ bits

Thus:

Higher entropy→greater tagging uncertainty Lower entropy→greater tagging confidence

The assessment specifically asks for entropy to be used to evaluate uncertainty **before and after rule application**.

Final Answers – Case Study 3

Question	Answer
Q1	economic/JJ growth/NN increases/VBZ employment/NN
Q2	increases/NNS is the initial error; it should be **VBZ** because “growth” is a singular NN subject
Q3	Total frequency = **1160**; probabilities: economic **0.1034**, growth **0.3879**, increases **0.1810**, employment **0.3276**
Q4	Brill Tagging corrects contextual POS errors through learned transformation rules; entropy measures uncertainty, with lower entropy indicating greater tagging confidence

Exam conclusion: Transformation-Based Tagging improves the initial POS output by applying **context-sensitive correction rules**. In this example, it changes **increases/NNS** to **increases/VBZ**, making the tag sequence grammatically consistent.